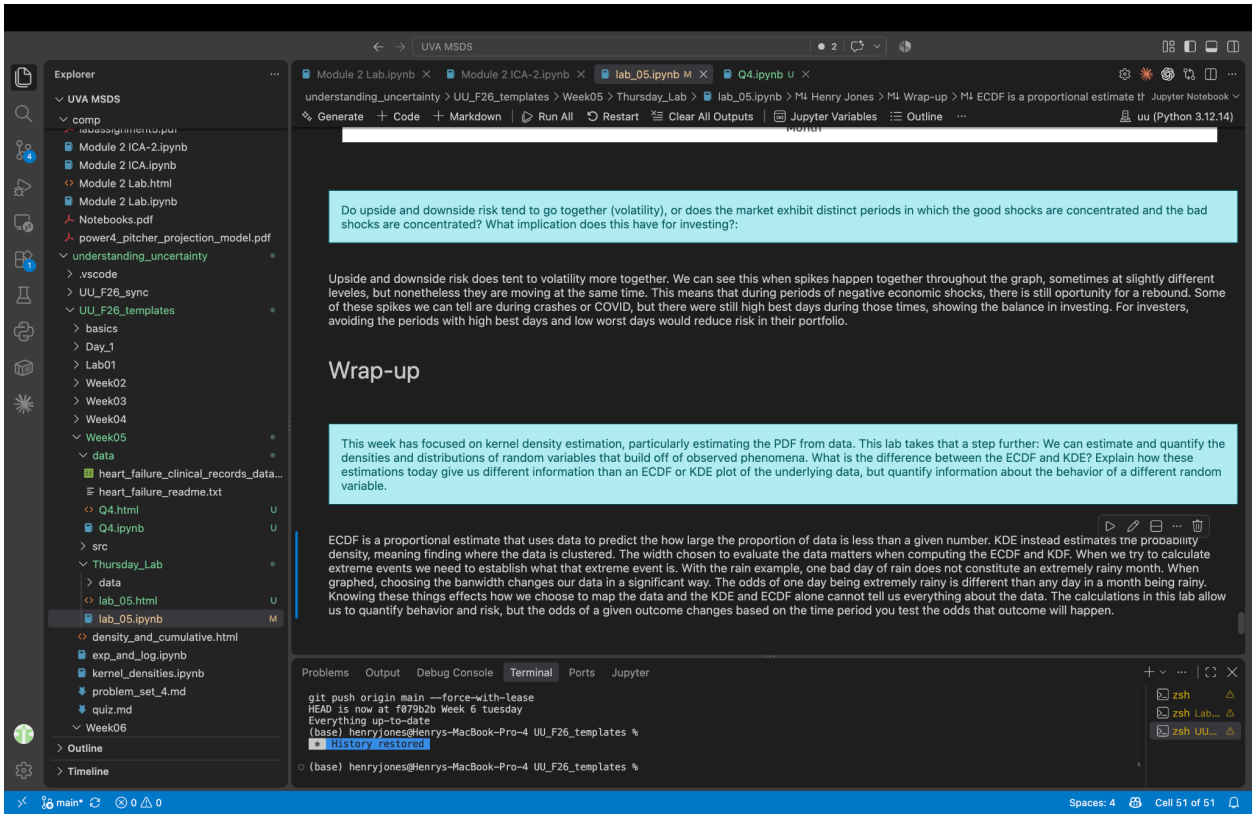
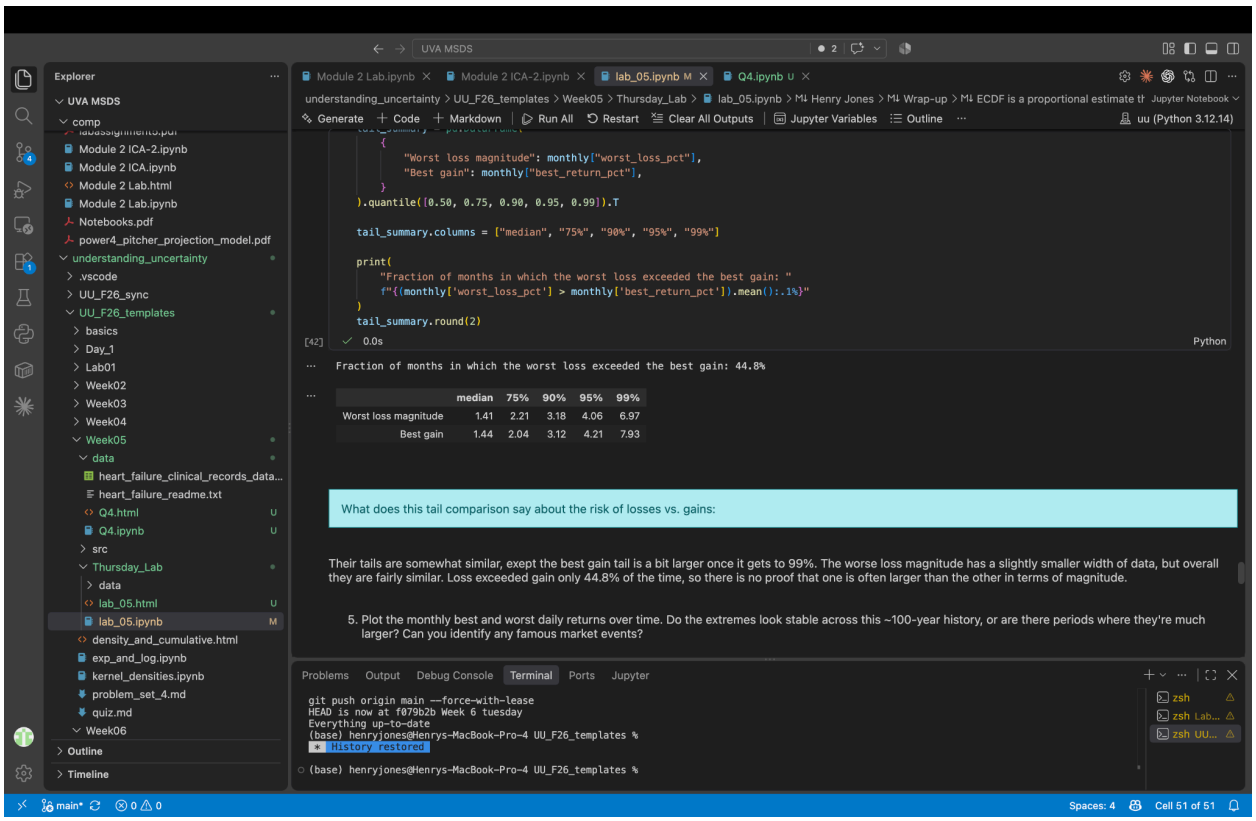
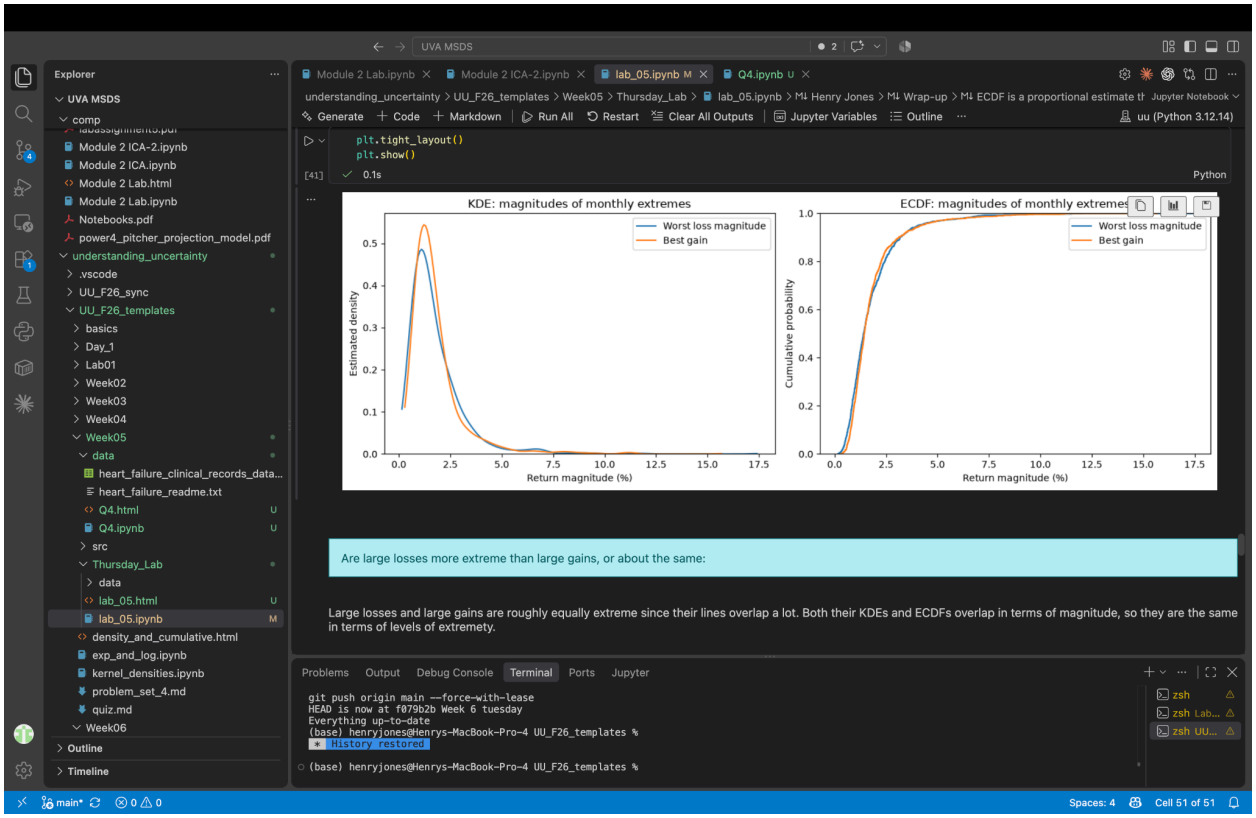
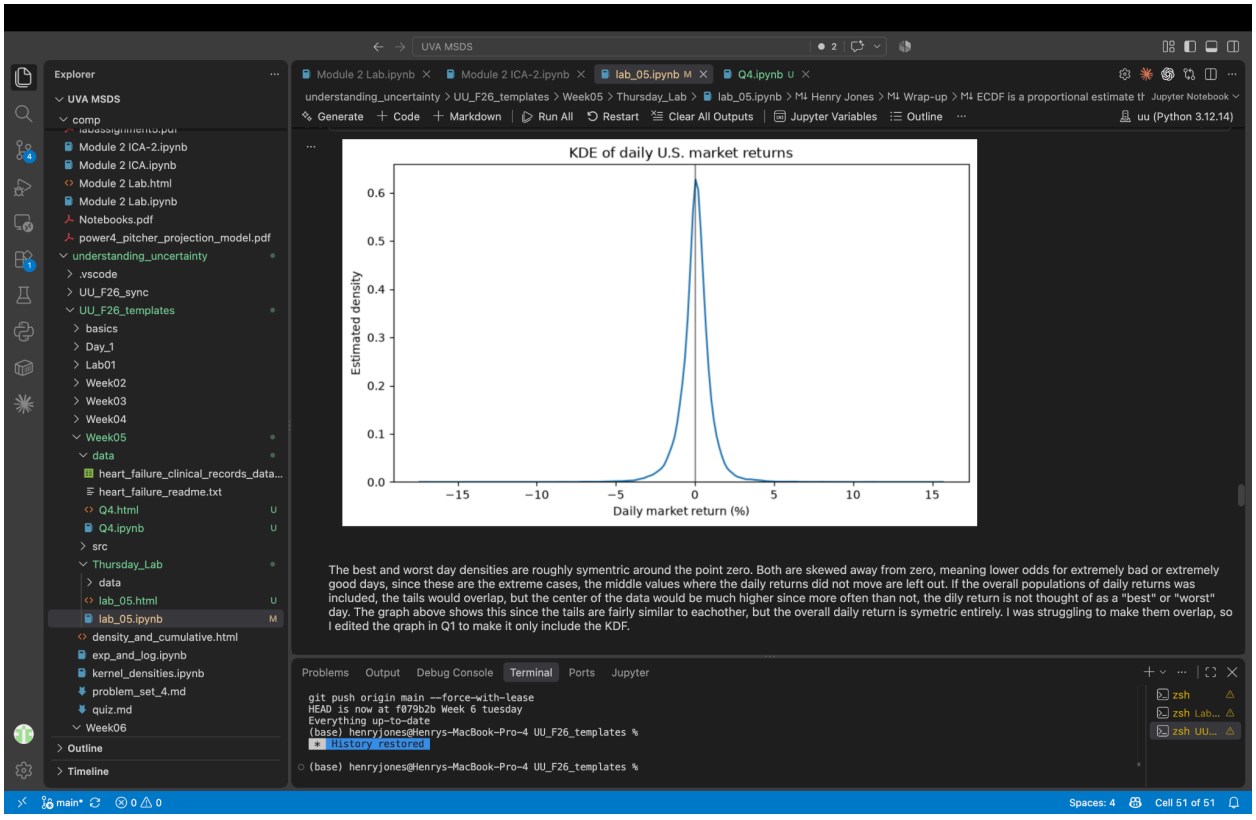
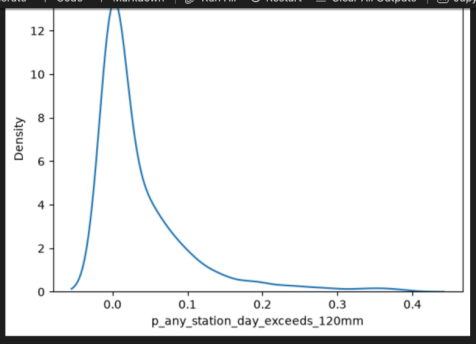

UVA MSDS
Explorer
UVA MSDS
comp
Module 2 ICA-2.ipynb
Module 2 ICA.ipynb
Module 2 Lab.html
Module 2 Lab.ipynb
Notebooks.pdf
power4_pitcher_projection_model.pdf
understanding_uncertainty
.vscode
UU_F26_sync
UU_F26_templates
basics
Day_1
Lab01
Week02
Week03
Week04
Week05
data
heart_failure_clinical_records_data...
heart_failure_readme.txt
Q4.html
Q4.ipynb
src
Thursday_Lab
data
lab_05.html
lab_05.ipynb
density_and_cumulative.html
exp_and_log.ipynb
kernel_densities.ipynb
problem_set_4.md
quiz.md
Week06
Outline
Timeline
Module 2 Lab.ipynb
Module 2 ICA-2.ipynb
lab_05.ipynb M
Q4.ipynb U
understanding_uncertainty > UU_F26_templates > Week05 > Thursday_Lab > lab_05.ipynb > Henry Jones > Wrap-up > ECDF is a proportional estimate th
Jupyter Notebook
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
Outline
uu (Python 3.12.14)
Do upside and downside risk tend to go together (volatility), or does the market exhibit distinct periods in which the good shocks are concentrated and the bad shocks are concentrated? What implication does this have for investing?:
Upside and downside risk does tent to volatility more together. We can see this when spikes happen together throughout the graph, sometimes at slightly different leveles, but nonetheless they are moving at the same time. This means that during periods of negative economic shocks, there is still oportunity for a rebound. Some of these spikes we can tell are during crashes or COVID, but there were still high best days during those times, showing the balance in investing. For investers, avoiding the periods with high best days and low worst days would reduce risk in their portfolio.
Wrap-up
This week has focused on kernel density estimation, particularly estimating the PDF from data. This lab takes that a step further: We can estimate and quantify the densities and distributions of random variables that build off of observed phenomena. What is the difference between the ECDF and KDE? Explain how these estimations today give us different information than an ECDF or KDE plot of the underlying data, but quantify information about the behavior of a different random variable.
ECDF is a proportional estimate that uses data to predict the how large the proportion of data is less than a given number. KDE instead estimates the probability density, meaning finding where the data is clustered. The width chosen to evaluate the data matters when computing the ECDF and KDF. When we try to calculate extreme events we need to establish what that extreme event is. With the rain example, one bad day of rain does not constitute an extremely rainy month. When graphed, choosing the banwidth changes our data in a significant way. The odds of one day being extremely rainy is different than any day in a month being rainy. Knowing these things effects how we choose to map the data and the KDE and ECDF alone cannot tell us everything about the data. The calculations in this lab allow us to quantify behavior and risk, but the odds of a given outcome changes based on the time period you test the odds that outcome will happen.
Problems
Output
Debug Console
Terminal
Ports
Jupyter
git push origin main --force-with-lease
HEAD is now at f079b2b Week 6 tuesday
Everything up-to-date
(base) henryjones@Henrys-MacBook-Pro-4 UU_F26_templates %
History restored
(base) henryjones@Henrys-MacBook-Pro-4 UU_F26_templates %
zsh
zsh Lab...
zsh UU...
main*
0 0
Spaces: 4
Cell 51 of 51


UVA MSDS
Explorer
UVA MSDS
comp
Module 2 ICA-2.ipynb
Module 2 ICA.ipynb
Module 2 Lab.html
Module 2 Lab.ipynb
Notebooks.pdf
power4_pitcher_projection_model.pdf
understanding_uncertainty
.vscode
UU_F26_sync
UU_F26_templates
basics
Day_1
Lab01
Week02
Week03
Week04
Week05
data
heart_failure_clinical_records_data...
heart_failure_readme.txt
Q4.html
Q4.ipynb
src
Thursday_Lab
data
lab_05.html
lab_05.ipynb
density_and_cumulative.html
exp_and_log.ipynb
kernel_densities.ipynb
problem_set_4.md
quiz.md
Week06
Outline
Timeline
Module 2 Lab.ipynb
Module 2 ICA-2.ipynb
lab_05.ipynb M
Q4.ipynb U
understanding_uncertainty > UU_F26_templates > Week05 > Thursday_Lab > lab_05.ipynb > Henry Jones > Wrap-up > ECDF is a proportional estimate th
Jupyter Notebook
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
Outline
uu (Python 3.12.14)
{
"Worst loss magnitude": monthly["worst_loss_pct"],
"Best gain": monthly["best_return_pct"],
}
).quantile([0.50, 0.75, 0.90, 0.95, 0.99]).T
tail_summary.columns = ["median", "75%", "90%", "95%", "99%"]
print(
"Fraction of months in which the worst loss exceeded the best gain: "
f"{(monthly['worst_loss_pct'] > monthly['best_return_pct']).mean():.1%}"
)
tail_summary.round(2)
[42]
0.0s
Python
Fraction of months in which the worst loss exceeded the best gain: 44.8%
median 75% 90% 95% 99%
Worst loss magnitude 1.41 2.21 3.18 4.06 6.97
Best gain 1.44 2.04 3.12 4.21 7.93
What does this tail comparison say about the risk of losses vs. gains:
Their tails are somewhat similar, exept the best gain tail is a bit larger once it gets to 99%. The worse loss magnitude has a slightly smaller width of data, but overall they are fairly similar. Loss exceeded gain only 44.8% of the time, so there is no proof that one is often larger than the other in terms of magnitude.
5. Plot the monthly best and worst daily returns over time. Do the extremes look stable across this ~100-year history, or are there periods where they're much larger? Can you identify any famous market events?
Problems
Output
Debug Console
Terminal
Ports
Jupyter
git push origin main --force-with-lease
HEAD is now at f079b2b Week 6 tuesday
Everything up-to-date
(base) henryjones@Henrys-MacBook-Pro-4 UU_F26_templates %
History restored
(base) henryjones@Henrys-MacBook-Pro-4 UU_F26_templates %
zsh
zsh Lab...
zsh UU...
main*
0 0
Spaces: 4
Cell 51 of 51


KDE: magnitudes of monthly extremes
Worst loss magnitude
Best gain
Estimated density
Return magnitude (%)
ECDF: magnitudes of monthly extremes
Worst loss magnitude
Best gain
Cumulative probability
Return magnitude (%)
Are large losses more extreme than large gains, or about the same:
Large losses and large gains are roughly equally extreme since their lines overlap a lot. Both their KDEs and ECDFs overlap in terms of magnitude, so they are the same in terms of levels of extremety.



The best and worst day densities are roughly symentric around the point zero. Both are skewed away from zero, meaning lower odds for extremely bad or extremely good days, since these are the extreme cases, the middle values where the daily returns did not move are left out. If the overall populations of daily returns was included, the tails would overlap, but the center of the data would be much higher since more often than not, the dily return is not thought of as a "best" or "worst" day. The graph above shows this since the tails are fairly similar to eachother, but the overall daily return is symetric entirely. I was struggling to make them overlap, so I edited the qraph in Q1 to make it only include the KDF.



Which states and months face the most risk of a greater-than-120mm day? How high is the probability? Among all state-months, describe the density of high precipitation days:

Hawaii in March has the greatesrt risk of an over 120mm day (0.387 probability). The spring months are also most likely to have a large rain month in these states. Overall, Hawaii, Florida, and Louisiana have the greatest chance of 120mm of rain in a month. Since most states have much lower probabilities of rain, the data is skewed right. Smaller number of states' months have a chance of having a month with 120mm of rain.